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AEROSOL DISPENSING APPARATUS

Abstract

A dispensing apparatus having a hollow body with a top opening and a bottom opening, the body including a discharge aperture therein, an actuator slidably arranged within the body and having a conduit therein, the conduit having a first opening and a second opening proximate a bottom thereof, a top cover secured to the body proximate the top, the cover having a movable section hingedly attached to the body, a leaf spring extending proximate the top opening and arranged to substantially abut a top surface of the movable section, thereby applying a downward force thereon, and a bottom cover removably secured to the bottom of the body, wherein the actuator is arranged to actuate an aerosol container within the body when downward force is 10 applied thereto, thereby applying force on a nozzle of the aerosol container and discharging therefrom through the conduit and out of the discharge aperture.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority pursuant to 35 U.S.C. 119(e) to U.S. Provisional Application No. 63/556,565, filed Feb. 22, 2024, which application is incorporated herein by reference in its entirety.

FIELD

[0002] The present invention relates to an apparatus for an aerosol canister, and more specifically, to a system for a safety lid mechanism arranged at a top of the apparatus and a ring-release system of the apparatus.

BACKGROUND

[0003] Cases for aerosol or spray canisters typically include an actuator to engage the stem of the canister in order to dispel the canister's contents. There are a variety of aerosol dispensers that are used in many applications which include dispensing perfume, air fresheners, personal hygiene products, covering an article with a coat of paint, and dispensing cleaning products, amongst others. One specific application for an aerosol dispenser is as a personal defense device that, for example, directs a chemical repellant spray towards a potential human or animal threat.

[0004] Many cases for aerosol or spray canisters are adapted for personal use, such that are smaller in size, allowing them to be stored in purses or small compartments. In some uses, these canisters have “ring-systems” allowing them to be attached to a key ring-typically affixed to the case.

[0005] It is also typical for aerosol or spray canisters to have lids for dispensing actuators or cases for aerosol and spray canisters that are intended to either prevent accidental discharge or provide easy dispensing. With respect to personal defense devices, easy dispensing is crucial to ensure the safety of the user when a threat presents itself. However, the reason personal defense devices are effective is because their chemical contents are indiscriminately extremely painful to anyone who comes into contact with it.

[0006] Typically, lids or caps, are movably affixed to aerosol or spray canisters vis-à-vis an axial pin and torsion style springs. Although reliable in function, this structural arrangement also increases the overall size, rather, limits potential overall size reductions for personal or handheld aerosol or spray canisters.

[0007] Therefore, there is a long-felt need for a personal or handheld aerosol or spray canister having a reduced overall size, a release system for a key ring attachment or other attachment, a leaf spring system made of singular pieces which maintains a cover over an actuator to prevent accidental discharge, and an internal sleeve having movable members which restrict translational movement of one or more of an actuator within the sleeve or an aerosol canister within the sleeve.

SUMMARY

[0008] In some aspects, the present invention may comprise a dispensing apparatus having a body having an open top section and a bottom section, further comprising, at least one dispensing aperture and a first attachment aperture proximate the top section, a second and third attachment aperture proximate the bottom section, a leaf spring hingedly attached to the top proximate the first attachment aperture, an actuator having a conduit defined by a dispensing opening and an inflow opening slidably arranged within the body, a top cover having a fixed section frictionally secured within the first attachment aperture and a movable section hingedly connected to the fixed section, a bottom cover rotatably secured to the bottom section of the body, and a quick-release attachment arranged to removably secure to the second and third attachment apertures, wherein an aerosol container is arranged to be removably secured within the body and dispenses contents from the dispensing aperture from the dispensing opening when the actuator is pushed towards the bottom

section within the sleeve.

[0009] In some embodiments, the aforementioned dispensing apparatus may further comprise a sleeve arranged within the body, the sleeve having an open top section and a bottom section, the leaf spring hingedly attached proximate the top section, the sleeve having a second dispensing aperture, the sleeve having a fourth attachment aperture proximate the top section, wherein the second dispensing aperture is arranged to be in communication with the at least one dispensing aperture and the fourth attachment aperture is arranged to be substantially colinear with the first attachment aperture of the body when the sleeve is arranged within the body, wherein the actuator and the aerosol container are arranged within the sleeve.

[0010] In other possible embodiments, the sleeve further comprises a first plurality of fingers and a second plurality of fingers, each of the fingers hingedly connected to the sleeve, wherein both of the plurality of fingers restrict downward slidable movement of the actuator within the sleeve.

[0011] In further contemplated configurations, the sleeve further comprises a first finger and a second being disposed on opposite sides of the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.

[0012] In some other possible embodiments, the body includes a first external guide rail and a second external guide rail disposed on an internal surface thereof, the external guide rails adapted to bias each of the fingers of the sleeve in a radially inward direction when the sleeve is within the body.

[0013] In other possible configurations the body of the dispensing apparatus further includes a first internal guide rail and a second internal guide rail disposed on an internal surface thereof and arranged between the external guide rails, the internal guide rails arranged to rest at least partially within a channel of the sleeve, thereby restricting downward movement of the sleeve within the body.

[0014] In some embodiments, the leaf spring biases the movable section of the top cover downwardly thereby maintaining the movable section above the actuator.

[0015] In some contemplated configurations the quick-release attachment comprises a key ring extending therefrom, the quick-release attachment having a pair of protrusions arranged to frictionally and removably engage the second and third attachment apertures.

[0016] In further possible embodiments, the quick-release attachment comprises a clip extending therefrom and arranged to rest on an outer surface of the body, the quick-release attachment having a pair of protrusions arranged to frictionally and removably engage the second and third attachment apertures.

[0017] In other aspects, the present invention may be a dispensing apparatus that includes a hollow body having a top opening and a bottom opening, the body including a discharge aperture proximate the top opening, an actuator slidably arranged within the body, the actuator having a conduit therein, the conduit having a first opening in communication with the discharge aperture and a second opening proximate the bottom, a top cover secured to the body proximate the top, the cover having a movable section hingedly attached to the body, a leaf spring extending from the top and arranged to substantially abut a top surface of the movable section, thereby applying a downward force thereon, and a bottom cover removably secured to the bottom, where the actuator is arranged to actuate an aerosol container within the body when downward force is applied to the actuator, thereby applying downward force to a nozzle of the aerosol container and discharging contents of the aerosol container through the conduit and out of the discharge aperture.

[0018] In other contemplated configurations, the leaf spring biases the movable section in a downward direction when the movable section is moved by a user in an upward direction, thereby substantially covering the actuator unless constant upward force is applied to the movable section.

[0019] In some possible embodiments, the aforementioned dispensing apparatus further comprises a quick-release attachment, the attachment having a pair of attachment tabs extending therefrom,

the attachment tabs removably and frictionally engaged to a respective pair of attachment apertures arranged proximate the bottom opening of the body.

[0020] In further possible designs, the quick release-attachment includes at least one of: a key-ring; an aperture formed by a portion extending from the body; and, a spring clip.

[0021] In some embodiments, the aforementioned dispensing apparatus recited in Claim **11** further comprises: a hollow sleeve having a top opening and bottom opening, the sleeve having a second discharge aperture therein, the sleeve including one of: at least a first plurality of fingers hingedly secured thereon, wherein the sleeve is arranged to be inserted within the body and the actuator is arranged to be inserted within the sleeve, wherein at least one finger of the first plurality of fingers restricts translational movement of the actuator in a direction of the bottom opening of the sleeve; or, a first finger and a second being disposed on opposite sides of and proximate the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.

[0022] In further possible embodiments, the body includes a cover aperture proximate the top opening thereof and the sleeve includes a cover aperture proximate the top opening thereof, wherein the top cover includes a fixed portion arranged to be frictionally secured at least partially within the cover apertures, thereby securing the sleeve within the body.

[0023] These and other objects, features, and advantages of the present invention will become readily apparent upon a review of the following detailed description of the invention, in view of the drawings and appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which:

[0025] FIGS. **1A** and **1B** illustrate perspective views of the present invention;

[0026] FIG. **1C** illustrates a top view of the invention shown in FIGS. **1A** and **1B**;

[0027] FIGS. **2A** and **2B** illustrate perspective views of body **100** of the present invention;

[0028] FIGS. **3A** and **3B** illustrate perspective views of cover **200** of the present invention;

[0029] FIGS. **4A** and **4B** illustrate perspective views of sleeve **300** of the present invention;

[0030] FIG. **4C** illustrates a partial front view of sleeve **300** of the present invention;

[0031] FIG. **5** illustrates a perspective view of bottom cover **400** of the present invention;

[0032] FIG. **6** illustrates a perspective view a quick-release attachment **500** of the present invention;

[0033] FIGS. **7A** and **7B** illustrate perspective views actuator **600** of the present invention;

[0034] FIG. **8A** illustrates a cross-sectional view taken generally along line A-A in FIG. **1C**;

[0035] FIG. **8B** illustrates a cross-sectional view taken generally along line B-B in FIG. **1C**;

[0036] FIGS. **9A** and **9B** illustrate exploded views of the present invention;

[0037] FIGS. **10A** and **10B** illustrate perspective views of the present invention having quick-release attachment **700** installed therein;

[0038] FIGS. **11A** and **11B** illustrate perspective views of quick-release attachment **700** of the present invention;

[0039] FIGS. **12A** and **12B** illustrate perspective views of a second embodiment of the present invention;

[0040] FIG. **13** illustrates a top view of the invention shown in FIGS. **12A** and **12B**;

[0041] FIGS. **14A** and **14B** illustrate perspective views of body **100'** of the invention shown in FIG. **12**;

[0042] FIGS. 15A through 15C illustrate sleeve 300' of the invention shown in FIGS. 12A and 12B;

[0043] FIG. 15D illustrates cover 200' within sleeve 300' of the invention shown in FIGS. 12A and 12B;

[0044] FIG. 16A illustrates a cross-sectional view taken generally along line AA-AA in FIG. 13;

[0045] FIG. 16B illustrates a cross-sectional view taken generally along line BB-BB in FIG. 13;

[0046] FIG. 17 illustrates a partial front perspective view of the invention shown in FIGS. 12A and 12B with a frontal portion thereof cutaway; and,

[0047] FIGS. 18A and 18B illustrate a partial cutaway of the invention shown in FIGS. 12A and 12B.

DETAILED DESCRIPTION

[0048] At the outset, it should be appreciated that like drawing numbers on different drawing views identify identical, or functionally similar, structural elements. It is to be understood that the claims are not limited to the disclosed aspects.

[0049] Furthermore, it is understood that this disclosure is not limited to the particular methodology, materials and modifications described and as such may, of course, vary. It is also understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to limit the scope of the claims. As such, those in the art will understand that any suitable material, now known or hereafter developed, may be used in forming the present invention described herein.

[0050] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure pertains. It should be understood that any methods, devices or materials similar or equivalent to those described herein or understood to one having skill in the art within, can be used in the practice or testing of the example embodiments.

[0051] The word "example" or "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any implementation described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other implementations.

[0052] If the specification states a component or feature "may," "can," "could," "should," "would," "preferably," "possibly," "typically," "optionally," "for example," "often," or "might" (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments, or it may be excluded.

[0053] The terms "having", "has", "including", "includes" are intended to be substantially synonymous with the terms "comprising" and/or "comprises" and/or "comprise".

[0054] It should be appreciated that the term "substantially" is synonymous with terms such as "nearly," "very nearly," "about," "approximately," "around," "bordering on," "close to," "essentially," "in the neighborhood of," "in the vicinity of," etc., and such terms may be used interchangeably as appearing in the specification and claims. It should be appreciated that the term "proximate" is synonymous with terms such as "nearby," "close," "adjacent," "neighboring," "immediate," "adjoining," etc., and such terms may be used interchangeably as appearing in the specification and claims. The term "approximately" is intended to mean values within ten percent of the specified value.

[0055] It should be understood that use of "or" in the present application is with respect to a "non-exclusive" arrangement, unless stated otherwise. For example, when saying that "item x is A or B," it is understood that this can mean one of the following: (1) item x is only one or the other of A and B; (2) item x is both A and B. Alternately stated, the word "or" is not used to define an "exclusive or" arrangement. For example, an "exclusive or" arrangement for the statement "item x is A or B" would require that x can be only one of A and B. Furthermore, as used herein, "and/or" is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions

recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element.

[0056] Moreover, as used herein, the phrases “comprises at least one of” and “comprising at least one of” in combination with a system or element is intended to mean that the system or element includes one or more of the elements listed after the phrase. For example, a device comprising at least one of: a first element; a second element; and, a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element. A similar interpretation is intended when the phrase “used in at least one of: ” or “one of: ” is used herein.

First Exemplary Embodiment

[0057] Adverting now to the figures, FIGS. **1A** and **1B** generally illustrate perspective view of the present invention, aerosol dispensing apparatus **10** (hereinafter “apparatus”), fully assembled and FIG. **1C** illustrates a top view of apparatus **10**.

[0058] As disclosed herein, apparatus generally comprises six (**6**) components, although it should be noted that various components may be combined in alternative configurations.

[0059] Generally, FIGS. **1A** through **1C** illustrate body **100**, top cover **200**, sleeve **300**, bottom cover **400**, quick-release attachment **500**. Within sleeve **300** and body **100** is actuator **600** (shown separated in FIGS. **7A** and **7B**). In operation, sleeve **300** also houses canister **12** (shown in FIGS. **9A** and **9B**), which is an canister arranged to have a pressured gas and/or vapor therein, such as but not limited to: pepper spray, mace, variations thereof (e.g., fogger, stream, gel, and foam embodiments), or other dispensable contents that are released from a canister upon the canister having pressure applied to an actuator.

[0060] The following description relates to possible individual components of apparatus **10** and are shown in FIGS. **2A** through **7B** individually.

[0061] FIGS. **2A** and **2B** generally illustrate body **100** of apparatus **10**. In some aspects, body **100** may include top **102** and bottom **104**, both of which are respective openings to internal space **106**. In other words, body **100** is hollow and includes two openings arranged at opposite ends thereof. In a preferred embodiment, discharge aperture **108** is arranged proximate top **102** and is open to internal space **106**. As shown in FIGS. **2A** and **2B**, discharge aperture **108** may be arranged within a section of body **100** that protrudes therefrom. First attachment aperture **110** and second attachment aperture **112** are preferably proximate bottom **104** and are open to internal space **106**. Like discharge aperture **108**, attachment apertures **110** and **112** may be arranged on respective sections of body **100** that protrude from first groove **120** and second groove **122**. Grooves **120** and **122** may be described as depressions arranged within an internal surface of body **100** and those depressions on the internal surface of body **100** protrude from the outer surface of the body **100**.

[0062] Similarly, third attachment aperture **114** may be arranged on the section of body **100** that protrudes therefrom and is positioned above discharge aperture **108** on the same protruding section, and attachment aperture **114** is open to internal space **106**. Body **100** may include first protrusion **116** and second protrusion **118** extending from an internal surface of body **100**, i.e., within internal space **106**. Protrusion **116** and **118** may have an arcuate arrangement and are substantially divided by attachment apertures **110** and **112** and/or grooves **120** and **122**.

[0063] In some embodiments, body **100** may have a substantially circular cross-section, however, it should be noted that various alternative geometries are contemplated, e.g., triangular, square, or any

other polygonal configuration, thus, elements of body **100** may have alternative configurations to accommodate different cross-sectional arrangements. In some configurations, the internal surface of body **100** may include shelves or abutments protruding into internal space **106**. These shelves or abutments may be respectively arranged above grooves **120** and **122**.

[0064] FIGS. **3A** and **3B** generally illustrate cover **200** of apparatus **10**. Cover **200** includes top **202** and bottom **204**, and hinge **206** which hingedly connects fixed section **208** and movable section **210**. It should be noted that “fixed” and “movable”, as used to designate the sections of cover **200** are in reference to their functional arrangement when apparatus is fully assembled and/or in use, and such designations should not be considered restrictive on the scope of the appended claims. Depression **212** is arranged within movable section **210** and generally proximate top **202**. First stop **214** and second stop **216** are protruding portions that extend from bottom **204** of fixed section **208**. Movable section **210** also includes first angled edge **218** and second angled edge **220** extending from a bottom thereof. Hinge **206** is preferably a living hinge, i.e., a flexure bearing, i.e., a thinner section of material made from the same material as the two “rigid” pieces it connects. In other words, sections **208** and **210** may pivot about hinge **206**. In a preferred arrangement, section **208** is fixed, therefore section **210** pivots about hinge **206** while section **208** remains stationary.

[0065] FIGS. **4A** and **4B** generally show sleeve **300** of apparatus **10**, and FIG. **4C** illustrates a partial front view of sleeve **300**. In some arrangements, sleeve **300** includes top **302** and bottom **304**, both of which generally are openings as sleeve **300** is a hollow body. Top **302** and bottom **304** define openings to internal space **306**. In a preferred embodiment, discharge aperture **308** is arranged proximate top **302** and is open to internal space **306**. Sleeve **300** includes first plurality of fingers **310** and second plurality of fingers **312** arranged proximate top **302**. Each finger of both plurality of fingers **310** and **312** are hingedly secured to sleeve **300** and form respective slots open to internal space **306** therebetween. Each finger of both plurality of fingers **310** and **312** may be hingedly secured to sleeve either proximate top **302** and/or bottom **304**, preferably in an alternating manner as generally depicted in FIGS. **4A** and **4B**. In a preferred configuration, each finger of both plurality of fingers **310** and **312** apply radially outward force in response to radially inward force applied thereon to maintain the position generally shown in FIGS. **4A** and **4B**. Thus, when the fingers are pushed inwardly, i.e., towards internal space, resistance from each of fingers is applied in the inverse direction. In a preferred configuration, sleeve **300** includes top surface **316** proximate top **302**. Leaf spring **320** is connected to a portion of top surface **316** via hinge **318**. As shown in FIG. **4C**, fourth attachment aperture **314** may have protrusions **326** and **328** arranged therein or at least proximate thereto. Protrusions **326** and **328** are tabs which may flex in directions towards top **302** or bottom **304**. Preferably, the protrusions will each have an angled face in the direction facing away from fourth attachment aperture **314** and a hard edge in a direction facing fourth attachment aperture **314**.

[0066] FIG. **5** illustrates bottom cover **400** of apparatus **10**. In some embodiments, cover **400** generally includes top **402** and bottom **404** having channel **406** disposed therein. First protrusion **408** and second protrusion **410** radially extend from cover **400** and are arranged proximate top **402**. First rim **414** and second rim **416** radially extend from cover **400** and are arranged proximate bottom **404**. First protrusion **408** and first rim **414** form channel **418** therebetween and similarly, second protrusion **410** and second rim **416** form channel **420** therein between. As shown in FIG. **5**, bottom cover **400** also includes key ring aperture **412** disposed therein.

[0067] FIG. **6** illustrates a quick-release attachment, e.g., key ring device **500**, of apparatus **10**. In some configurations, key ring device **500** includes body **502** having first extension **504** and second extension **506** extending upwardly therefrom, where first extension **504** has first protruding tab **508** hingedly secured thereto and second extension **506** has first protruding tab **510** hingedly secured thereto. Tabs **508** and **510** arranged such that respective portions thereof extend radially outward past extensions **504** and **506** and apply radially outward force in response to radially inward force applied thereon to maintain the position generally shown in FIG. **6**. In some arrangements, body

502 may include a protruding portion extending therefrom, e.g., housing **512**. Housing **512** includes aperture **514** therein. In some embodiments, housing **512** is also arranged to accept keyring **516** therein via slot **518**, where slot **518** extends into housing **512** allowing keyring **516** may rest within housing **512**. Further, keyring **516** may have a portion thereof extending upwardly from slot **518** and away from body **502**.

[0068] FIGS. 7A and 7B illustrate perspective views of actuator **600** of apparatus **10**. Actuator **600** includes top **602**, bottom **606**, top surface **616**, sidewall **618**, and conduit **606** arranged therein. Conduit **606** is defined by two openings, dispensing opening **608** arranged within sidewall **618** and inflow opening **610** proximate bottom **606**. Dispensing opening **608** is in fluid communication with inflow opening **610**. In some configurations, sidewall **618** includes first plurality of slots **612** and second plurality of slots **614** arranged therein.

[0069] In some embodiments and generally shown in FIGS. 10A through 11B, apparatus **10** may comprise quick-release attachment **700**. Generally, quick-release attachment **700** may comprise all of the components of quick-release attachment **500** with the addition of a clip, allowing apparatus **10** to be removably attached to belt loop and the like. Quick-release attachment **700** includes body **702** having clip extension **720** extending therefrom. Clip body **722** is connected to clip extension **720** via curved portion **724**. Clip body **722** is bounded by curved portion **724** and end **726**. As shown in FIGS. 10A and 10B, clip body **722** rests on an outer surface of body **100** and functions like a known-in-the-art spring clip, i.e., clip body **722** applies constant force onto the surface of body **100**, allowing an object to be sandwiched therebetween. It should be noted that in alternative embodiments, quick-release attachment **700** may not include housing **712** and therefore aperture **714** with key-ring therein.

[0070] The following description should be taken in view of all of the aforementioned Figures, and specifically, FIGS. 8A through 9B. FIG. 8A illustrates a cross-sectional view of apparatus **10** taken generally along line A-A in FIG. 1C and FIG. 8B illustrates a cross-sectional view of apparatus **10** taken generally along line B-B in FIG. 1C. FIGS. 9A and 9B generally show exploded perspective views of apparatus **10**.

[0071] To assemble apparatus **10**, first actuator **600** is slid into internal space **306** of sleeve **300** such that actuator **600** is arranged therein proximate top **302**. Then sleeve **300** is slid into internal space **106** of body **100**. When sleeve **300** is inserted into internal space **106** a surface thereof applies inward radial force onto each finger of both first plurality of fingers **310** and second plurality of fingers **312**, such that the fingers are forcibly bent into internal space **306** of sleeve **300**. This action allows each of the respective fingers of first plurality of fingers **310** and second plurality of fingers **312** to at least partially nest within first plurality slots **612** and second plurality of slots **614** of actuator **600**, thereby maintaining actuator within internal space **306** of sleeve **300**. At least one finger of first plurality of fingers **310** and/or second plurality of fingers **312** will abut at least one slot of first plurality slots **612** and/or second plurality of slots **614**, thereby limiting the distance actuator **600** may slide within internal space **306** of sleeve **300** in a direction towards bottom **304** thereof. Thus, the combination of first plurality of fingers **310** and/or second plurality of fingers **312**, when bent inwardly into internal space **306**, and first plurality slots **612** and second plurality of slots **614** functions as a translational distance restrictor of actuator **600**.

[0072] Top cover **200** maintains sleeve **300** within body **100**. First, sleeve **300** is slid within internal space **106** of body **100** until third attachment aperture **114** of body **100** and fourth attachment aperture **314** of sleeve **300** are substantially aligned, i.e., substantially colinear, thus discharge aperture **108** of body **100** and second discharge aperture **308** of sleeve **300** are also substantially aligned, i.e., substantially colinear. Fixed section **208** of top cover **200** is inserted into both third attachment aperture **114** of body **100** and fourth attachment aperture **314** of sleeve **300**. In some embodiments, one or more of fourth attachment aperture **314** or third attachment aperture **114** includes at least one tab therein. As shown in FIG. 4C, in a preferred embodiment, fourth attachment aperture **314** includes protrusions **326** and **328** (i.e., tabs). Fixed section, or portion, **208**

is inserted into attachment apertures **114** and **314** and protrusions **326** and **328** are flexed in a downward direction from stops **214** and **216** of fixed section **208** pushing into the angled faces of protrusions **326** and **328**. Once fixed section is inserted far enough within at least attachment aperture **314**, stops **214** and **216** no longer apply pressure on protrusions **326** and **328** allowing protrusions to return to an unflexed position where the hard edge of the protrusions abuts stops **214** and **216**, preventing fixed section **208** from being removed from attachment aperture **314** and therefore also preventing fixed section **208** from being removed from attachment aperture **114**.

[0073] Once top cover **200** is installed within sleeve **300** and body **100**, leaf spring **320** rests within depression **212** of movable section, or portion, **210** of top cover **200**. Leaf spring **320** rests, i.e., does not apply force, when positioned at least partially within or on depression **212**. When movable section **210** is lifted, i.e., pivoted about hinge **206** in a direction towards top **102** of body **100**, leaf spring **320** applies force onto depression **212** thereby pivoting movable section **210** in a direction towards bottom **104** of body **100**. This action keeps movable section **210**, thereby cover **200**, over actuator **600**, so that accidental downward pressure on actuator **600** cannot be applied (i.e., preventing accidental discharge of the contents of canister **12**). Additionally, angled edges **218** and **220** of movable section **210** are arranged to abut edge **330** of sleeve **300** to prevent unwanted downward moveable (i.e., (i.e., preventing accidental discharge of the contents of canister **12**).

[0074] Similarly, movable section **210** needs force applied in the upward direction towards top **102** in order to expose top surface **616** of actuator **600**. When movable section **210** is hinged and maintained in the upward position, force may be applied in a direction towards bottom **104** of body **100** on top surface of actuator **600**, thereby applying downward force on nozzle **14** of canister **12**. When nozzle **14** is depressed downwardly, i.e., towards bottom **104**, pressurized contents are release therefrom, up into inflow opening **610** of conduit **606** and out of dispensing opening **608**, discharge aperture **308** and discharge aperture **108**. So long as downward pressure is applied to top surface **616** of actuator **600**, pressurized contents from canister **12** will be dispensed until canister is emptied and/or depressurized.

[0075] At this point, body **100**, sleeve **300**, actuator **600**, and cover **200** are assembled and canister **12** may be inserted within internal space **306** of sleeve **300** within body **100**. Canister **12** is slid into internal space **306** until nozzle **14** is at least partially within inflow opening **610** of actuator **600**. In some arrangements, at least one finger of first plurality of fingers **310** and second plurality of fingers **312**, when bent radially inward within internal space **306** of sleeve **300**, may contact a top portion of canister **12** to prevent canister **12** from sliding further in a direction towards top **302**. Since nozzle **14** of canister **12** is arranged to dispense upon downward force applied thereon, the at least one finger prevents accidental discharge of nozzle **14** by preventing canister **14** from sliding further in a direction towards top **302**. Unless bottom cover **600** is engaged to bottom **104** of body **100**, canister **12** may freely slide out of internal space **306** of sleeve **300** proximate bottom **104**.

[0076] Bottom cover **400** rotatably secured to bottom **102** of body **100**. Within internal space **106** proximate bottom **102** is first protrusion **116** and second protrusion **118**, both of which separated by first groove **120** and second groove **122**. In some aspects, bottom **102**, or the bottom end, of body **100** has a pair of lips partially circumscribed therearound and disposed substantially between grooves **120** and **122**, thusly forming a pair of arcuate channels below protrusions **116** and **118** and between the lip of bottom **102**. Thus, protrusions **408** and **410** of bottom cover **400** are first inserted into grooves **120** and **122**. Then bottom cover **400** is rotated about 90° so that protrusions **408** and **410** are positioned within each of the arcuate channels and below protrusion **116** and **118**. Bottom cover **400** thereby prevents canister **12** from sliding through bottom **304** of sleeve **300** and bottom **104** of body **100**.

[0077] To finalize the assembly of apparatus **10**, quick-release attachment **500** is removably attached to body **100** proximate bottom **104**. Body **502** is first aligned within channel **406** of bottom cover **400** and key-ring **516** is aligned with aperture **412**. Then protruding tabs **508** and **510** of extensions **504** and **506** are inserted into grooves **120** and **122** of body **100** proximate bottom

104, thereby bending tabs **508** and **510** towards one another until tabs **508** and **510** are seated within first and second attachment apertures **110** and **112** of body **100**, i.e., in a snap-fit and/or quick-release manner, allowing tabs **508** and **510** to release in a radial outward direction. To remove quick-release attachment **500**, tabs **508** and **510** are pushed/bent towards one another so they are released from their respective attachment apertures, allowing quick-release attachment **500** to be removed from body **100**. It should be noted that quick-release attachment **700** removably engages body **100** of apparatus **10** in substantially identical fashion as quick-release attachment **500**.

Second Exemplary Embodiment

[0078] Adverting now to FIGS. **12** through **20B** which generally illustrate a second embodiment of apparatus **10**, e.g., apparatus **10'**. At the outset, the notable difference between the embodiments is the configuration of the respective sleeves, e.g., sleeve **300** of apparatus **10** and sleeve **300'** of apparatus **10'**, specifically, the selection of either a pair of a plurality of fingers or a pair of spaced fingers, e.g., plurality of fingers **310**, **312** of apparatus **10** and fingers **332**, **334** of apparatus **10'**.

[0079] The following description should be taken in view of FIGS. **12A** through **13**. FIGS. **12A** and **12B** illustrate perspective views of a second embodiment of a dispensing apparatus, designated generally as **10'**, with top cover **200'** in a closed position. Apparatus **10'** comprises a body **100'**, top cover **200'**, a sleeve (not shown), and bottom cover **400'**. Body **100'** includes a dispensing aperture proximate top **102'**. A first attachment aperture and second attachment aperture (not shown) are located proximate bottom **104'** for receiving quick release attachment **500'** which secures bottom cover **400'** to body **100'**. Top cover **200'** is hingedly attached to body **100'**. Arranged proximate bottom **104'** of body **100'** is first protrusion **116'** and second protrusion **118'**, substantially similar to those of body **100** in FIG. **2B** and FIG. **8B**, which protrusions interact with bottom cover **400'** (shown in FIG. **12B**) to removably secure the bottom cover to body **100'**. A notable difference between the protrusions of body **100** and those of body **100'** is that first protrusion **116'** and second protrusion **118'** include vertical extensions **116A** and **118A** that protrude from the internal surface of the sidewall of body **100'**. Vertical extensions **116A** and **118A** restrict vertical movement of a second sleeve within body **100'**, which sleeve is shown in FIGS. **16A** and **16B**.

[0080] FIG. **13** illustrates a top view of dispensing apparatus **10'**. Top cover **200'** includes fixed section **208'** and movable section **210'**, hingedly connected at hinge **206'**. Movable section **210'** includes depression **212'** which is designed to accept leaf spring **320'** therein, where leaf spring **320'** (of the sleeve within body **100'**) biases movable section **210'** of top cover **200'** in a downward direction. Fixed section **208'** is designed to be inserted within body **100'**, thereby connecting top cover **200'** to body **100'**.

[0081] FIGS. **14A** and **14B** illustrate perspective views of body **100'** of dispensing apparatus **10'**, where FIG. **14B** is partially cutaway. Body **100'** is generally cylindrical and includes an open top **102'** and an open bottom **104'**, defining openings to an internal space **106'**. Dispensing aperture **108'** is positioned proximate to top **102'**, providing an opening for dispensing the contents of an aerosol container (not shown). A first attachment aperture **114'** is also located proximate to top **102'** and is configured to receive a portion of a top cover (not shown), specifically, a fixed section of the top cover. Proximate bottom **104'** are first attachment aperture **110'** and second attachment aperture **112'**, configured to receive a quick-release attachment (not shown).

[0082] Within internal space **106'**, a pair of internal guide rails **128** and **130** extend from a curved portion **124** towards top **102'**. These guide rails are positioned below first attachment aperture **114'** and are configured to interface with a sleeve (not shown) that houses the aerosol container and actuator (neither shown). A pair of external guide rails, **126** and **132**, also extend inwardly from curved portion **124** towards top **102'**. These external guide rails are positioned above first attachment aperture **110'** and second attachment aperture **112'**, respectively, and are configured to interface with fingers (not shown) of the sleeve. External guide rails **126** and **132** are longer than internal guide rails **128** and **130**. Curved portion **124** transitions smoothly and defines a larger

diameter portion of internal space **106'** proximate top **102** as compared to the portion of body **100'** proximate bottom **104'** and beneath curved portion **124**. First protrusion **116'** and second protrusion **118'** extend inwardly from bottom **104'** in a circumferential manner. These protrusions aid in securing bottom cover **400'** (not shown) to body **100'**.

[0083] FIGS. **15A** through **15C** illustrate sleeve **300'** of dispensing apparatus **10'**. FIG. **15A** illustrates a perspective view of sleeve **300'**, highlighting the three-dimensional relationship between leaf spring **320'**, fingers **332** and **334**, and channel **342**. FIG. **15B** illustrates another perspective view of sleeve **300'**, rotated relative to the view shown in FIG. **15A**. This view further emphasizes the three-dimensional shape of sleeve **300'** and the arrangement of its features. FIG. **15C** illustrates a front view of sleeve **300'**, providing a clearer depiction of protrusions **326'** and **328'** extending downward from an internal surface of attachment aperture **314'**, as well as the living hinges connecting fingers **332** and **334** to wall **330'** adjacent to channel **342**.

[0084] Sleeve **300'** is generally cylindrical and includes an open top **302'** and open bottom **304'**, which define openings to internal space **306'**. A circumferential wall defines sleeve **300'** extends between and connects top **302'** and bottom **304'** and has edge **330'** arranged opposite from dispensing aperture **308'**. Sleeve **300'** further includes dispensing aperture **308'**, attachment aperture **314'**, leaf spring **320'**, first finger **332**, second finger **334**, and channel **342**. Dispensing aperture **308'** is proximate top **302'** and provides a passage for dispensing the contents of an aerosol container (not shown).

[0085] Attachment aperture **314'** is proximate top **302'** and includes protrusions **326'** and **328'** that extend downward, as best shown in FIG. **15C**. These protrusions are configured to interact with a top cover (not shown), substantially identical those present in apparatus **10**, described supra. Leaf spring **320'** is proximate top **302'** and is hingedly attached to sleeve **300'** via hinge **318'**, which is preferably a living hinge. The free end of leaf spring **320'** extends over top **302'**. First finger **332** and second finger **334** are located dispensing aperture **308'**. Each finger, **332** and **334**, is hingedly attached to wall **330'** by respective living hinges. The living hinges allow fingers **332** and **334** to flex inwardly towards internal space **306'**. Channel **342** is disposed on the exterior surface of wall **330'**, opposite dispensing aperture **308'**. This channel is configured to interact with guide rails (not shown) on the body (not shown) of the dispensing apparatus.

[0086] With reference to FIG. **15A** through **15D**, FIG. **15D** illustrates a top plan view of actuator **600'** positioned within sleeve **300'**. Actuator **600'** is generally cylindrical and includes top surface **616'**, sidewall **618'**, medial guide depression **620**, first side guide depression **622**, and second side guide depression **624**. Medial guide depression **620** is centrally located and arranged within a respective external surface, while first side guide depression **622** and second side guide depression **624** are positioned on opposite sides of medial guide depression **620**. These depressions are configured to interact with corresponding protrusions on sleeve **300'** to maintain alignment during translation in the vertical direction and restrict horizontal/rotational movement of the actuator within the sleeve.

[0087] Sleeve **300'** in FIG. **15D** shows edge **330'**, leaf spring **320'**, first side guide protrusion **338**, second side guide protrusion **340**, and medial guide protrusion **336**. Top surface **316'** is shown and has living hinge **318'** of leaf spring **320'**. Leaf spring **320'** is positioned substantially between top surface **316'** (proximate the dispensing aperture) and edge **330'** and is configured to bias a top cover (not shown) in a closed position. First side guide protrusion **338** and second side guide protrusion **340** extend radially inward and are positioned to interface with the corresponding side guide depressions, **624** and **622**, on actuator **600'**. Medial guide protrusion **336** extends inwardly from edge **330'** and is positioned to interface with medial guide depression **620** of actuator **600'**. This arrangement of protrusions and depressions ensures that actuator **600'** remains centered within sleeve **300'** during vertical translational movement and also restricts horizontal/rotational movement within sleeve **300'**.

[0088] Comparing sleeve **300** (shown in FIGS. **4A** through **4C**) with sleeve **300'** (shown in FIGS.

15A through 15C) reveals a key difference in the number and arrangement of the fingers that interact with the actuator. This design difference impacts how each sleeve interacts with the corresponding actuator (**600** and **600'**, respectively).

[0089] Sleeve **300** includes first plurality of fingers **310** and second plurality of fingers **312**. These fingers are arranged circumferentially around the sleeve proximate the top opening and are configured to flex inwardly when the sleeve is inserted into body **100** of dispensing apparatus **100**. The plurality of fingers, **310** and **312**, on sleeve **300** are designed to interact with the multiple slots on actuator **600**.

[0090] In contrast, sleeve **300'** includes only two fingers, **332** and **334**, located on opposite sides of the sleeve proximate dispensing aperture **308'** and within sidewall **301'**. These fingers are each hingedly attached to sidewall **301'** by a living hinge. This simplified design is intended simplify the manufacturing process and reduce the potential of breakage. Additionally, sleeve **300'** is designed to interact with medial guide depression **620**, side guide depression **622** and second side guide depression **624** of actuator **600'** via its corresponding protrusions (**336**, **338**, and **340**). This configuration with fewer interacting elements reduces complexity while providing precise alignment and controlled movement of the actuator within the sleeve.

[0091] Comparing actuator **600** (shown in FIGS. 3A and 3B) with actuator **600'** (shown in FIG. 15D) reveals key differences in their structure and interaction with other components of the dispensing apparatus. Actuator **600** is designed to interact with a sleeve (**300**) having a plurality of fingers, while actuator **600'** interacts with sleeve **300'** having only two fingers (**332**, **334**). This difference is also reflected in the design of the guide depressions on actuator **600'**.

[0092] Actuator **600** includes first plurality of slots **612** and second plurality of slots **614** arranged circumferentially around its sidewall (**618**). These slots are configured to receive and interact with the plurality of fingers of sleeve **300**, restricting downward movement of the actuator.

[0093] In contrast, actuator **600'** includes medial guide depression **620**, first side guide depression **622**, and second side guide depression **624**. These depressions interface with medial guide protrusion **336**, first side guide protrusion **338**, and second side guide protrusion **340** of sleeve **300'**, respectively. This configuration with fewer, more precisely positioned interacting elements allows for controlled movement of actuator **600'** within sleeve **300'** while still providing stability. It also accommodates the different design of sleeve **300'**, which has only two fingers (**332**, **334**) instead of a plurality of fingers. This design reduces complexity while maintaining functionality.

[0094] FIG. 16A is a cross-sectional view taken generally along line AA-AA in FIG. 13 and FIG. 16B is a cross-sectional view taken generally along line BB-BB in FIG. 13. Both figures illustrate dispensing apparatus **10'**. Apparatus **10'** comprises body **100'**, top cover **200'**, sleeve **300'**, actuator **600'**, bottom cover **400'**, and quick release attachment **500'**. Apparatus **10'** is designed to accommodate canister **12'** therein.

[0095] In FIG. 16A, body **100'** includes a third attachment aperture **114'** proximate a top of body **100'**, a first attachment aperture **110'** and a second attachment aperture **112'** proximate a bottom of body **100'**, as generally shown in FIG. 14A. Top cover **200'** includes fixed section **208'** within third attachment aperture **114'** and attachment aperture **314'** of sleeve **300'**. Top cover **200'** also includes movable section **210'** hinged connected to fixed section **208'**. Sleeve **300'** is nested within body **100'**. Lower sleeve **800** is nested within body **100'** and below sleeve **300'**. Actuator **600'** includes conduit **610'** and dispensing opening **608'**. Canister **12** includes nozzle **14**. Bottom cover **400'** is removably attached to body **100'** proximate bottom **104'** of body **100'**. Quick release attachment **500'** includes first extension **504'**, second extension **506'**, and body **502'**.

[0096] FIG. 16B illustrates a similar view as in FIG. 16A, however, the cross-section cuts through first attachment aperture **110'** proximate bottom **104'** and second protrusion **118'** of body **100'**. First protrusion **116'** is also shown. The other elements shown are the same as described in FIG. 16A, however, fewer features are shown due to the location of the cross-section.

[0097] It should be noted that bottom cover **400'** may also include the components of bottom cover

400, as shown in FIG. 5. It should also be noted that quick-release attachment **500'** is substantially similar to quick-release attachment **500** shown in FIG. 6 but it designed to omit ring **516**. It should be further noted that apparatus **10'** could removably engage attachment **700**, generally shown in FIGS. **10A** through **11B**, or alternatively, quick-release attachment **500'** could be configured with some or all of the additional components of attachment **700**.

[0098] FIG. **17** provides a partial front perspective view of the dispensing apparatus **10'**, with a portion cutaway to reveal the internal components and their interactions. The apparatus comprises body **100'**, top cover **200'**, sleeve **300'**, and actuator **600'**. A portion of bottom cover **400'** and quick release attachment **500'** (See FIGS. **12A** and **12B**) would be visible below body **100'** in this view, but are omitted for clarity and because they do not directly interact with the components shown.

[0099] Body **100'** is partially sectioned to show internal guide rails **128** and **130** and external guide rails **126** and **132**. Internal guide rails **128** and **130** extend from curved portion **124** of the body and from an internal surface thereof and are configured to fit within channel **342** of sleeve **300'**.

External guide rails **126** and **132** are positioned out of the internal guide rails and extend higher up on the inner wall of body **100'** and extends inwards, designed to interact with and bias first finger **332** and second finger **334** of sleeve **300'** inwardly. The positioning of these guide rails helps to secure and align sleeve **300'** within body **100'**. Internal guide rails **130** and **128** are also visible within body **100'**. These guide rails are shorter than guide rails **126** and **132**, and work in conjunction with the other guide rails to stabilize, center, and restrict the vertical movement within the body of the internal components, e.g., actuator **600'** and sleeve **300'**.

[0100] Sleeve **300'** is shown nested within body **100'**. Channel **342**, located near the bottom opening (**304'**, not shown) of the sleeve, receives internal guide rails **130** and **128**, further securing the sleeve's vertical position within the body and also preventing rotational movement with the body. First finger **332** and second finger **334** extend inwardly from the sidewall of sleeve **300'** via external guide rails **132** and **126** and are each hingedly connected to the sleeve wall by living hinges, allowing them flexibility. These fingers are positioned close to the dispensing opening **608'** of actuator **600'**. The interaction of fingers **332** and **334** with external guide rails **132** and **126**, creates a stop which restricts vertical movement of actuator **600'**, controlling the dispensing action. Leaf spring **320'**, attached to the upper portion of sleeve **300'**, extends upwards and is configured to interact with top cover **200'**, at least partially nested within depression **212'**.

[0101] Top cover **200'** includes fixed section **208'** and movable section **210'**. Movable section **210'** includes depression **212'**, which receives at least a portion of leaf spring **320'**. The leaf spring biases movable section **210'** downwards, towards body **100'**, keeping top cover **200'** in a closed position and until the user intentionally opens it. Top cover **200'** is hingedly attached to body **100'** at the top.

[0102] Actuator **600'** is nested within sleeve **300'** and its dispensing opening **608'** is aligned with the dispensing aperture (**108'**, not shown) of body **100'** and dispensing opening **308'** of sleeve **300'**. When top cover **200'** is open (shown in FIG. **18A** and **18B** and described further, infra), downward pressure on actuator **600'** causes an aerosol container within sleeve **300'** to dispense its contents. The cutaway view reveals the internal workings of the apparatus and how the various components interact to control the dispensing process.

[0103] FIGS. **18A** and **18B** offer partial cutaway views of dispensing apparatus **10'**, providing further detail on the arrangement and interaction of internal components, building upon the view shown in FIG. **17**. Apparatus **10'** comprises body **100'**, top cover **200'**, sleeve **300'**, lower sleeve **800**, and actuator **600'**. These figures highlight the interplay between the body's guide rails and the sleeve components, particularly the fingers of sleeve **300'**, where the interplay restricts vertical translational movement of actuator **600'** within sleeve **300'** and body **100'**.

[0104] In FIG. **18A**, a portion of body **100'**, top cover **200'** and sleeve **300'** is cut away to reveal finger **332** and sleeve **300'**. Sleeve **300'** is nested within body **100'**. As shown in FIG. **17**, first finger **332** and second finger **334**, integral parts of sleeve **300'**, are biased inward by the interaction

with external guide rails **132** and **126**. This inward bias ensures that the fingers maintain contact with actuator **600'**, specifically shelf **626**, further restricting its vertical movement when actuator **600'** is pushed in a direction away from top cover **200'**. Lower sleeve **800** is nested within body **100'**, adding a layer of stability and support for actuator **600'** and also providing a housing for canister **10** therein. Leaf spring **320'**, attached to sleeve **300'** near its top, extends upward and engages with depression **212'** in top cover **200'**. FIGS. **18A** and **18B** illustrate a substantially "open" position of top cover **200'**, specifically when upward force is imparted on bottom **204'** of top cover **200'**. Top cover **200'**, comprising fixed section **208'** and movable section **210'**, is biased in closed position by leaf spring **320'**. The angled edges (**218'**, **220'**) of movable section **210'** may contact edge **330'** of sleeve **300'** when the cover is closed.

[0105] FIG. **18B** presents a similar view as FIG. **18A**, but with a different section of body **100'** removed to reveal external guide rail **132**. This alternate cutaway provides a comprehensive view of the external guide rails, e.g., **132**, bias the fingers, e.g., **332**, radially inward within body **100'**. The inclusion of shelf **626** on actuator **600'** further restricts the actuator's downward movement within lower sleeve **800**, sleeve **300'**, and body **100'**, as shelf **626** is designed to contact or abut one or more of the fingers of sleeve **300'**.

[0106] It should be noted that the disclosed embodiments may be arranged in various combinations according to any of the embodiments shown and described. As such, the illustrated and described embodiments are intended to be illustrative and various alternatives, combinations, omissions, of specific components, or foreseeable alternative components, understood by one having ordinary skill in the art, described in the present disclosure or within the field of the present disclosure, are intended to fall within the scope of the appending claims.

[0107] It will be appreciated that various aspects of the invention and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

LIST OF REFERENCE NUMERALS

[0108] **10** Apparatus [0109] **10'** Apparatus [0110] **12** Canister [0111] **14** Nozzle [0112] **100** Body [0113] **100'** Body [0114] **102** Top [0115] **102'** Top [0116] **104** Bottom [0117] **104'** Bottom [0118] **106** Internal space [0119] **106'** Internal space [0120] **108** Discharge aperture [0121] **108'** Discharge aperture [0122] **110** First attachment aperture [0123] **110'** First attachment aperture [0124] **112** Second attachment aperture [0125] **112'** Second attachment aperture [0126] **114** Third attachment aperture [0127] **114'** Third attachment aperture [0128] **116** First protrusion [0129] **116'** First protrusion [0130] **118** Second protrusion [0131] **118'** Second protrusion [0132] **120** First groove [0133] **122** Second groove [0134] **124** Curved portion [0135] **126** First outer guide rail [0136] **128** First internal guide rail [0137] **130** Second internal guide rail [0138] **132** Second external guide rail [0139] **200** Top cover [0140] **200'** Top cover [0141] **202** Top [0142] **204** Bottom [0143] **204'** Bottom [0144] **206** Hinge [0145] **206'** Hinge [0146] **208** Fixed section [0147] **208'** Fixed section [0148] **210** Movable section [0149] **210'** Movable section [0150] **212** Depression [0151] **212'** Depression [0152] **214** First stop [0153] **216** Second stop [0154] **218** First angled edge [0155] **218'** First angled edge [0156] **220** Second angled edge [0157] **220'** Second angled edge [0158] **300** Sleeve [0159] **300'** Sleeve [0160] **301** Sidewall [0161] **301'** Sidewall [0162] **302** Top [0163] **302'** Top [0164] **304** Bottom [0165] **304'** Bottom [0166] **306** Internal space [0167] **306'** Internal space [0168] **308** Second discharge aperture [0169] **308'** Second discharge aperture [0170] **310** First plurality of fingers [0171] **312** Second plurality of fingers [0172] **314** Fourth attachment aperture [0173] **314'** Attachment aperture [0174] **316** Top surface [0175] **316'** Top surface [0176] **318** Hinge [0177] **318'** Hinge [0178] **320** Leaf spring [0179] **302'** Leaf spring [0180] **322** First extension [0181] **324** Second extension [0182] **326** Protrusion [0183] **326'** Protrusion [0184] **328** Protrusion [0185] **328'** Protrusion [0186] **330** Edge [0187] **330'** Edge [0188] **332** First finger [0189] **334**

Second finger [0190] **336** Medial guide protrusion [0191] **338** First side guide protrusion [0192]
340 Second side guide protrusion [0193] **342** Channel [0194] **400** Bottom cover [0195] **400'**
 Bottom cover [0196] **402** Top [0197] **404** Bottom [0198] **406** Channel [0199] **408** First protrusion
 [0200] **410** Second protrusion [0201] **412** Aperture [0202] **414** First rim [0203] **416** Second rim
 [0204] **418** First channel [0205] **420** Second channel [0206] **500** Quick-release attachment [0207]
500' Quick-release attachment [0208] **502** Body [0209] **502'** Body [0210] **504** First extension
 [0211] **504'** First extension [0212] **506** Second extension [0213] **506'** Second extension [0214] **508**
 First protruding tab [0215] **508'** First protruding tab [0216] **510** Second protruding tab [0217] **510'**
 Second protruding tab [0218] **512** Housing [0219] **512'** Housing [0220] **514** Aperture [0221] **514'**
 Aperture [0222] **516** Keyring [0223] **518** Slot [0224] **600** Actuator [0225] **600'** Actuator [0226] **602**
 Top [0227] **602'** Top [0228] **604** Bottom [0229] **604'** Bottom [0230] **606** Conduit [0231] **606'**
 Conduit [0232] **608** Dispensing opening [0233] **608'** Dispensing opening [0234] **610** Inflow
 opening [0235] **610'** Inflow opening [0236] **612** First plurality of slots [0237] **614** Second plurality
 of slots [0238] **616** Top surface [0239] **616'** Top surface [0240] **618** Sidewall [0241] **618'** Sidewall
 [0242] **620** Medial guide depression [0243] **622** First side guide depression [0244] **624** Second side
 guide depression [0245] **626** Shelf [0246] **700** Quick-release attachment [0247] **702** Body [0248]
704 First extension [0249] **706** Second extension [0250] **708** First protruding tab [0251] **710**
 Second protruding tab [0252] **712** Housing [0253] **714** Aperture [0254] **718** Slot [0255] **720** Clip
 extension [0256] **722** Clip body [0257] **724** Arcuate portion [0258] **726** End [0259] **728** Surface
 [0260] **800** Lower sleeve

Claims

1. A dispensing apparatus comprising: a body having an open top section and a bottom section, further comprising: at least one dispensing aperture and a first attachment aperture proximate the top section; a second and third attachment aperture proximate the bottom section; a leaf spring hingedly attached to the top proximate the first attachment aperture; an actuator having a conduit defined by a dispensing opening and an inflow opening slidably arranged within the body; a top cover having a fixed section frictionally secured within the first attachment aperture and a movable section hingedly connected to the fixed section; a bottom cover rotatably secured to the bottom section of the body; and, a quick-release attachment arranged to removably secure to the second and third attachment apertures; wherein an aerosol container is arranged to be removably secured within the body and dispenses contents from the dispensing aperture from the dispensing opening when the actuator is pushed towards the bottom section within the sleeve.
2. The dispensing apparatus recited in claim 1 further comprising: a sleeve arranged within the body, the sleeve having an open top section and a bottom section, the leaf spring hingedly attached proximate the top section, the sleeve having a second dispensing aperture, the sleeve having a fourth attachment aperture proximate the top section, wherein the second dispensing aperture is arranged to be in communication with the at least one dispensing aperture and the fourth attachment aperture is arranged to be substantially colinear with the first attachment aperture of the body when the sleeve is arranged within the body, wherein the actuator and the aerosol container are arranged within the sleeve.
3. The dispensing apparatus recited in claim 2, wherein the sleeve further comprises a first plurality of fingers and a second plurality of fingers, each of the fingers hingedly connected to the sleeve, wherein both of the plurality of fingers restrict downward slidable movement of the actuator within the sleeve.
4. The dispensing apparatus recited in claim 2, wherein the sleeve further comprises a first finger and a second being disposed on opposite sides of the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.

5. The dispensing apparatus recited in claim 4, wherein the body includes a first external guide rail and a second external guide rail disposed on an internal surface thereof, the external guide rails adapted to bias each of the fingers of the sleeve in a radially inward direction when the sleeve is within the body.
6. The dispensing apparatus recited in claim 5, wherein the body further includes a first internal guide rail and a second internal guide rail disposed on an internal surface thereof and arranged between the external guide rails, the internal guide rails arranged to rest at least partially within a channel of the sleeve, thereby restricting downward movement of the sleeve within the body.
7. The dispensing apparatus recited in claim 1, wherein the leaf spring biases the movable section of the top cover downwardly thereby maintaining the movable section above the actuator.
8. The dispensing apparatus recited in claim 1, wherein the quick-release attachment comprises a key ring extending therefrom, the quick-release attachment having a pair of protrusions arranged to frictionally and removably engage the second and third attachment apertures.
9. The dispensing apparatus recited in claim 1, wherein the quick-release attachment comprises a clip extending therefrom and arranged to rest on an outer surface of the body, the quick-release attachment having a pair of protrusions arranged to frictionally and removably engage the second and third attachment apertures.
10. A dispensing apparatus, comprising: a hollow body having a top opening and a bottom opening, the body including a discharge aperture proximate the top opening; an actuator slidably arranged within the body, the actuator having a conduit therein, the conduit having a first opening in communication with the discharge aperture and a second opening proximate the bottom; a top cover secured to the body proximate the top, the cover having a movable section hingedly attached to the body; a leaf spring extending from the top and arranged to substantially abut a top surface of the movable section, thereby applying a downward force thereon; and, a bottom cover removably secured to the bottom; wherein the actuator is arranged to actuate an aerosol container within the body when downward force is applied to the actuator, thereby applying downward force to a nozzle of the aerosol container and discharging contents of the aerosol container through the conduit and out of the discharge aperture.
11. The dispensing apparatus recited in claim 10, wherein the leaf spring biases the movable section in a downward direction when the movable section is moved by a user in an upward direction, thereby substantially covering the actuator unless constant upward force is applied to the movable section.
12. The dispensing apparatus recited in claim 10 further comprising: a quick-release attachment, the attachment having a pair of attachment tabs extending therefrom, the attachment tabs removably and frictionally engaged to a respective pair of attachment apertures arranged proximate the bottom opening of the body.
13. The dispensing apparatus recited in claim 12, wherein the quick release-attachment includes at least one of: a key-ring; and, a spring clip.
14. The dispensing apparatus recited in claim 11 further comprising: a hollow sleeve having a top opening and bottom opening, the sleeve having a second discharge aperture therein, the sleeve including one of: at least a first plurality of fingers hingedly secured thereon, wherein the sleeve is arranged to be inserted within the body and the actuator is arranged to be inserted within the sleeve, wherein at least one finger of the first plurality of fingers restricts translational movement of the actuator in a direction of the bottom opening of the sleeve; or, a first finger and a second being disposed on opposite sides of and proximate the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.
15. The dispensing apparatus recited in claim 14, wherein the body includes a cover aperture proximate the top opening thereof and the sleeve includes a cover aperture proximate the top

opening thereof, wherein the top cover includes a fixed portion arranged to be frictionally secured at least partially within the cover apertures, thereby securing the sleeve within the body.
